DELHI-NCR CAMPUS, GHAZIABAD (U.P)

ASSIGNMENT PROJECT

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Statistical machine learning(18CSE479T)

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MOTIVASION

In today's rapidly evolving digital landscape, the sheer volume of information generated poses a significant challenge for individuals seeking to stay informed and up-to-date on relevant topics. Traditional methods of consuming information, such as meticulously reading lengthy articles or diligently watching extended videos, can be time-consuming and inefficient. To address this challenge, text summarization has emerged as a promising solution, offering the ability to condense lengthy pieces of text into concise and informative summaries. By automatically generating summaries of text, text summarization tools empower individuals to quickly grasp the key points of a document and make informed decisions.

The pervasive presence of text in various forms, including news articles, research papers, and social media posts, highlights the practical applications of text summarization. In the context of news articles, text summarization can enable users to quickly scan headlines and identify relevant articles, saving valuable time and effort. Similarly, researchers can leverage text summarization to swiftly review lengthy research papers, gaining an overview of key findings and methodologies. Moreover, social media users can benefit from text summarization to extract the essence of lengthy posts, facilitating efficient information consumption.

ABSTRACT

This project aims to develop a robust and efficient text summarization system capable of generating concise and informative summaries of news articles. The system will employ an extractive summarization approach, meticulously selecting the most important sentences from the original text to construct a comprehensive summary. To identify the most salient sentences, the TF-IDF (Term Frequency-Inverse Document Frequency) algorithm will be utilized, a statistical measure that reflects the importance of a word in a document relative to the entire corpus.

The system will be meticulously trained on a vast corpus of news articles and rigorously evaluated using standard metrics such as ROUGE (Recall-Oriented Understudy for Gisting Evaluation) and BLEU (Bilingual Evaluation Understudy) to ensure its effectiveness and accuracy.

TECHNIQUES USED

To achieve the objective of generating informative and concise summaries, the system will employ a combination of Natural Language Processing (NLP) techniques and machine learning algorithms. NLP techniques, such as tokenization, stemming, and stop word removal, will be employed to preprocess the text, transforming it into a structured format suitable for analysis. Subsequently, the TF-IDF algorithm will be utilized to rank the importance of sentences based on their word frequencies and distribution across the dataset.

Finally, the system will apply an extractive summarization algorithm to select the most important sentences based on their rankings, effectively constructing a concise and informative summary.

ALGORITHMS

The system will leverage several algorithms to achieve its text summarization goals:

TF-IDF (Term Frequency-Inverse Document Frequency):

This statistical measure quantifies the importance of a word in a document by considering its frequency within that document and its rarity across the entire corpus.

Sentence Ranking Algorithm:

A method for sorting sentences based on their importance scores, effectively identifying the most salient sentences within the text.

Extractive Summarization Algorithm:

A technique for selecting the most important sentences to form the summary, ensuring that the key points of the original text are captured concisely.

By effectively utilizing these algorithms, the system will generate summaries that not only capture the essence of the original text but also maintain coherence and readability, providing users with a valuable tool for efficient information consumption.

CODE

```python
In [ ]: import nltk
from nltk.corpus import stopwords
from nltk.tokenize import sent_tokenize

def text_summarization(text):
    # Preprocess the text
    stop_words = set(stopwords.words('english'))
    sentences = sent_tokenize(text)
    processed_sentences = []

    for sentence in sentences:
        words = [word.lower() for word in sentence.split() if word not in stop_words]
        processed_sentences.append(' '.join(words))

    # Calculate sentence importance using TF-IDF
    from sklearn.feature_extraction.text import TfidfVectorizer
    vectorizer = TfidfVectorizer()
    sentence_tfidf = vectorizer.fit_transform(processed_sentences)
    sentence_scores = np.array(sentence_tfidf.sum(axis=1)).ravel()

    # Select top sentences based on importance scores
    num_sentences = int(0.2 * len(sentences))  # Adjust for desired summary length
    top_indices = sentence_scores.argsort()[-num_sentences:]
    top_sentences = [processed_sentences[i] for i in top_indices]

    # Generate the summary
    summary = ' '.join(top_sentences)
    return summary

# Example usage
text = "The quick brown fox jumps over the lazy dog. The dog barks at the fox. The fox runs away."
summary = text_summarization(text)
print(summary)
```

OUTPUT

```
quick brown fox jumps lazy dog. dog barks fox. fox runs away.
```